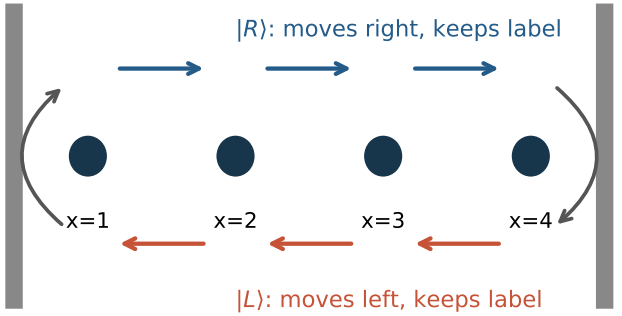
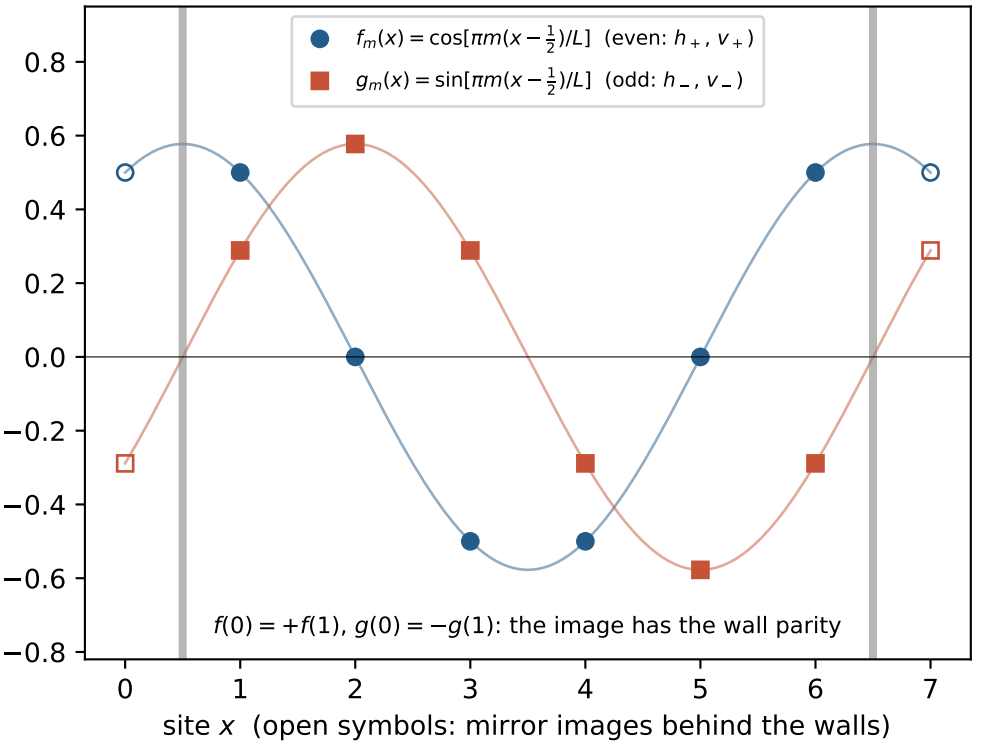
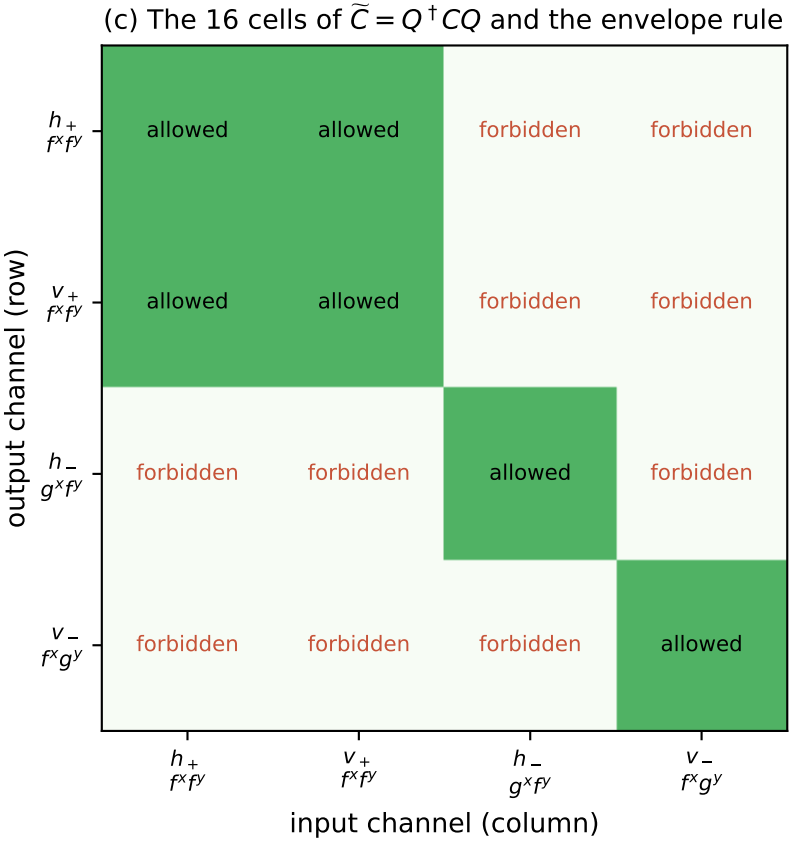
(a) The wall reverses the direction; $h_\pm$, $v_\pm$ are its ±1 eigenvectors



blocked move:
stay, $R \to L$
(amplitude +1)

At a vertical wall the shift acts on the pair $\{R, L\}$ as the reversal J:
$J|h_+\rangle = +|h_+\rangle$, $J|h_-\rangle = -|h_-\rangle$, $h_\pm = (R \pm L)/\sqrt{2}$.
Same at horizontal walls for $\{U, D\}$ and $v_\pm = (U \pm D)/\sqrt{2}$.

(b) Standing-wave envelopes on a segment ($L = 6$, $m = 2$)



(c) The 16 cells of $\widetilde{C} = Q^\dagger C Q$ and the envelope rule